Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/16/2009
PI Name:	Bedford, Joel Ph.D.		
Project Title:	Dose-Rate Effects and Components of Systems Governing Variations in Susceptibility for Carcinogenic and Acute Radiation Risks following Gamma-Ray, Proton, or HZE Irradiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80523	**Congressional District:**	4
Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Task Description:	There are two hypotheses. First, dose-rate effects in the 1 to 0.005 Gy/min range will be the same for protons vs. gamma-rays for chromosomal aberration induction. Second, levels of gamma-H2AX foci in our assays will parallel differences in chromosomal radiosensitivity, and with differences in carcinogenic and acute effects radiosensitivity among individuals or different strains of mice, known from other studies. Tracking will also apply for protons and perhaps to a lesser extent for HZE particles. Such a biomarker may help identify differences among individuals in a component contributing to their susceptibility to these effects. The aims are: 1) to compare dose-rate effects for gamma-rays vs. protons for chromosomal aberration induction in cultured human fibroblast and epithelial cells. (This would test assumptions regarding equivalence of gamma-ray and high energy protons with respect to RBEs and dose-rate effects of interest for astronauts.) 2) to determine optimal conditions for use of gamma-H2AX as a biomarker of DNA damage status related to aberration induction, for comparison with the aim 1 results, and to maximize differences in the assay for cells with radiosensitivities known to range from mild to severe; (These comparisons would serve as a preliminary step toward linking these damage endpoints to carcinogenesis and cell killing and measuring dose-rate effects for protons in a variety of cells.) 3) to determine whether changes in the 2D culture or 3D tissue-like context of cells may affect the utility of the putative markers for predicting carcinogenic and acute effects radiosensitivity; and 4) to compare the effect of dose-rate biomarker responses for protons vs gamma-rays in a G0 gamma-H2AX assay and for HDR using a G2/M assay in cells from mice with known differences in carcinogenic radiosensitivity or in human cells where such differences are known. Some comparisons with HZE particles would also be included.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of variation among individuals in their susceptibility to radiation carcinogenesis would be important information, not only of interest to individual astronauts who will be exposed to radiation during space missions, but would also be of interest to anyone who may be exposed to radiation in their occupation or in connection with activities or experiences where radiation exposure may be involved.
Task Progress:	One aim of this research is to answer a practical question on whether dose-rate effects for protons from solar particle events (SPE) that might be experienced by astronauts, are similar or perhaps the same as dose-rate effects for gamma-rays. While the effect of interest is carcinogenesis, the question is being addressed using the surrogate damage end-point of chromosomal aberrations in low passage cultured normal human cells. This would help indicate the possible necessity for carrying out more complicated animal carcinogenesis proton dose-rate experiments. A second aim involves development of a biomarker that may be predictive of the extent of variation in radiation sensitivity among individuals. However, it is first important to determine whether the assertion that normal cells may respond differently in 2D vs 3D culture is true for the biomarkers we are studying, viz., G2 chromosomal radiosensitivity and dose rate sensitivity with respect to the steady state levels of gamma-H2AX foci in cells at the end of a low dose rate radiation exposure. Dose-Rate Effects for 50 Mev and 1 Gev Protons vs. Gamma-rays: Addressing the first major aim required preliminary studies to determine the extent of dose rate effects for chromosomal aberration induction in the cell system we are using and most importantly, to determine whether this dose rate effect can be resolved for gamma-rays for dose-rates that could reasonably be expected to be provided for protons at either NSRL at Brookhaven National Laboratory. Last year we reported that dose-rate effects for chromosomal aberration induction were clearly distinguishable for total doses of 2 and 4 Gy delivered at 0.48 Gy/h compared to 150 Gy/h. We have now secured a 2-day time slot an NSRL to compare dose rate effects for chromosomal aberrations in 2 normal human cells using 50 Mev protons vs gamma-rays over a dose range of 1 to 3 Gy where the low dose rate will be 0.3 Gy/h and the high dose rate will be around 50 Gy/h. On the second day the same conditions (cells, doses, dose rates, etc.) will apply, but the proton energy will be 1 Gev. The gamma-ray parallel controls cannot be entirely carried out at NSRL, but a high dose rate dose response will be carried out along with the parallel low dose rate series in our gamma-radiation facility at CSU. The high dose rate gamma-ray dose response will serve to determine whether the shipping of cells to NSRL and back had any influence on the result. Cells in 2D Monolayer vs. 3D Ductal Structures We have developed the MCF10A mammary epithelial cell system used extensively by Bissell and colleagues, and Barcellos-Hoff and colleagues to facilitate study of cellular responses in a culture system that more closely simulates the “tissue context” of cells. These cells can be grown in monolayer 2D culture, much like any other cell in the traditional or classical cell culture situation, or the same cells can be grown in matrigel with certain growth factors, such that the cells form ductal acinar structures that closely simulate milk-producing mammary ducts. Since it has been strongly asserted that cells in such 3D tissue like contexts “respond very differently than in a 2D monolayer contest” we have extended our preliminary experiments reported last year to compare the formation and disappearance of gamma-H2AX after irradiation for the same cells in the 2D vs 3D condition. Our results have now established that for chromosome aberration induction, cell survival, and the levels of gamma-H2AX after a 20 hour continuous irradiation at 0.10 Gy/h the 2D monolayer vs. 3D acinar or ductal growth condition have absolutely no influence on these responses for the MCF10A ductal epithelial cells. These studies were important for our project since it indicates that we do not need to switch entirely to 3D culture systems for our development of a biomarker of radiosensitivity. In other words it appears that cell responses in 2D culture are not likely to be so misleading as some have predicted.
Bibliography Type:	Description: (Last Updated: 02/04/2013)
Articles in Peer-reviewed Journals	Lin Y-F, Nagasawa H, Peng Y, Chuang EY, Bedford JS. "Comparison of Several Cellular Radiation Effects in Human MCF10A Mammary Epithelial Cells Cultured as 2D Monolayers or 3D Acinar Structures in Matrigel." Radiation Research (In Press, 2009). , Apr-2009